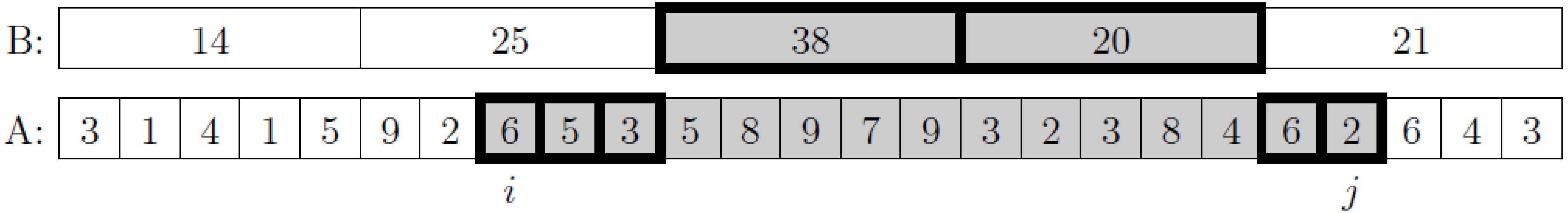
Five Nice Ideas…
1. Macro / Micro Decomposition

In a sequence, support two operations:

- Modify an element
- Find the sum of any range



- Easy to achieve O(√n) time for each using an auxiliary array of block sums (√n blocks of size √n). Crude but simple.
- Would also work for things like range min / max.

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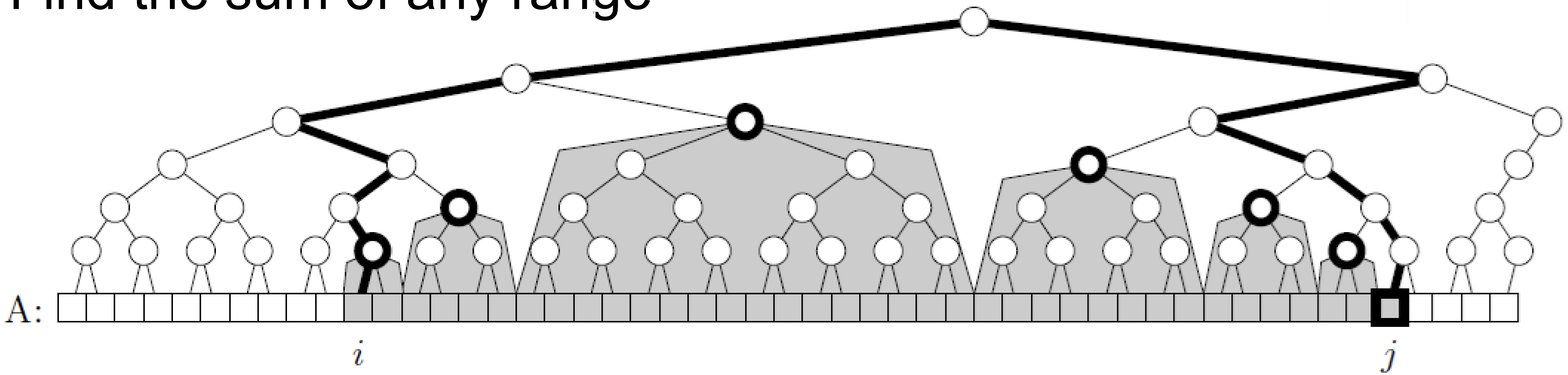

- Generalizes to a "segment tree", a static complete binary tree built "on top of" our array (often level-order encoded in an array like a binary heap). Very similar to a leaf-oriented BST.